

```
In [49]: import pandas as pd
```

The history saving thread hit an unexpected error (OperationalError('attempt to write a readonly database')).History will not be written to the database.

```
In [50]: ratings = pd.read_csv(r'/Users/abhiramireddygali/Downloads/march(2025).csv')
```

```
In [51]: ratings.shape
```

```
Out[51]: (20000263, 4)
```

```
In [52]: movies = pd.read_csv(r'/Users/abhiramireddygali/Downloads/march(2025).csv')
```

```
In [53]: movies.shape
```

```
Out[53]: (27278, 3)
```

```
In [54]: tags = pd.read_csv(r'/Users/abhiramireddygali/Downloads/march(2025).csv')
```

```
In [55]: tags.shape
```

```
Out[55]: (465564, 4)
```

```
In [56]: movies.movieId
```

```
Out[56]: 0          1
         1          2
         2          3
         3          4
         4          5
         ...
        27273    131254
        27274    131256
        27275    131258
        27276    131260
        27277    131262
        Name: movieId, Length: 27278, dtype: int64
```

```
In [57]: movies.head(20)
```

Out [57]:

	movieid	title	genres
0	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
1	2	Jumanji (1995)	Adventure Children Fantasy
2	3	Grumpier Old Men (1995)	Comedy Romance
3	4	Waiting to Exhale (1995)	Comedy Drama Romance
4	5	Father of the Bride Part II (1995)	Comedy
5	6	Heat (1995)	Action Crime Thriller
6	7	Sabrina (1995)	Comedy Romance
7	8	Tom and Huck (1995)	Adventure Children
8	9	Sudden Death (1995)	Action
9	10	GoldenEye (1995)	Action Adventure Thriller
10	11	American President, The (1995)	Comedy Drama Romance
11	12	Dracula: Dead and Loving It (1995)	Comedy Horror
12	13	Balto (1995)	Adventure Animation Children
13	14	Nixon (1995)	Drama
14	15	Cutthroat Island (1995)	Action Adventure Romance
15	16	Casino (1995)	Crime Drama
16	17	Sense and Sensibility (1995)	Drama Romance
17	18	Four Rooms (1995)	Comedy
18	19	Ace Ventura: When Nature Calls (1995)	Comedy
19	20	Money Train (1995)	Action Comedy Crime Drama Thriller

```
In [58]: del ratings['timestamp']
del tags['timestamp']
```

```
In [59]: ratings.columns
```

```
Out[59]: Index(['userId', 'movieId', 'rating'], dtype='object')
```

```
In [60]: tags.columns
```

```
Out[60]: Index(['userId', 'movieId', 'tag'], dtype='object')
```

Data structures

```
In [61]: row_0 = tags.iloc[0] # to get specific row  
         type(row_0)
```

```
Out[61]: pandas.core.series.Series
```

```
In [62]: row_0
```

```
Out[62]: userId          18  
movieId         4141  
tag      Mark Waters  
Name: 0, dtype: object
```

```
In [63]: row_0.index
```

```
Out[63]: Index(['userId', 'movieId', 'tag'], dtype='object')
```

```
In [64]: row_0['userId']
```

```
Out[64]: np.int64(18)
```

```
In [65]: 'rating' in row_0
```

```
Out[65]: False
```

```
In [66]: row_0.name
```

```
Out[66]: 0
```

```
In [67]: row_0 = row_0.rename('First row')  
         row_0.name
```

```
Out[67]: 'First row'
```

Data Frames

```
In [68]: tags.head()
```

Out[68]:

	userId	movieId	tag
--	--------	---------	-----

0	18	4141	Mark Waters
1	65	208	dark hero
2	65	353	dark hero
3	65	521	noir thriller
4	65	592	dark hero

In [69]: `tags.index`

Out[69]: `RangeIndex(start=0, stop=465564, step=1)`

In [70]: `tags.columns`

Out[70]: `Index(['userId', 'movieId', 'tag'], dtype='object')`

In [71]: `tags.iloc[[0,11,500]]`

Out[71]:

	userId	movieId	tag
0	18	4141	Mark Waters
11	65	1783	noir thriller
500	342	55908	entirely dialogue

Descriptive statistics

In [73]: `ratings['rating'].describe()`

Out[73]:

count	2.000026e+07
mean	3.525529e+00
std	1.051989e+00
min	5.000000e-01
25%	3.000000e+00
50%	3.500000e+00
75%	4.000000e+00
max	5.000000e+00
Name: rating, dtype: float64	

In [74]: `ratings.describe()`

Out[74]:

	userId	movieId	rating
count	2.000026e+07	2.000026e+07	2.000026e+07
mean	6.904587e+04	9.041567e+03	3.525529e+00
std	4.003863e+04	1.978948e+04	1.051989e+00
min	1.000000e+00	1.000000e+00	5.000000e-01
25%	3.439500e+04	9.020000e+02	3.000000e+00
50%	6.914100e+04	2.167000e+03	3.500000e+00
75%	1.036370e+05	4.770000e+03	4.000000e+00
max	1.384930e+05	1.312620e+05	5.000000e+00

In [75]: ratings['rating'].mean()

Out[75]: np.float64(3.5255285642993797)

In [76]: ratings.mean()

Out[76]:

userId	69045.872583
movieId	9041.567330
rating	3.525529
dtype:	float64

In [77]: ratings['rating'].min()

Out[77]: 0.5

In [78]: ratings['rating'].max()

Out[78]: 5.0

In [79]: ratings['rating'].std()

Out[79]: 1.051988919275684

In [80]: ratings['rating'].mode()

Out[80]:

0	4.0
Name: rating, dtype: float64	

In [81]: ratings.corr()

Out[81]:

	userId	movieId	rating
userId	1.000000	-0.000850	0.001175
movieId	-0.000850	1.000000	0.002606
rating	0.001175	0.002606	1.000000

```
In [82]: filter1 = ratings['rating'] > 10
         print(filter1)
         filter1.any()

0         False
1         False
2         False
3         False
4         False
...
20000258  False
20000259  False
20000260  False
20000261  False
20000262  False
Name: rating, Length: 20000263, dtype: bool

Out[82]: np.False_
```

```
In [84]: filter2 = ratings['rating'] > 0
         filter2.all()
```

```
Out[84]: np.True_
```

Data cleaning: Handling missing data

```
In [86]: movies.shape
```

```
Out[86]: (27278, 3)
```

```
In [89]: movies.isnull().any().any()
```

```
Out[89]: np.False_
```

```
In [90]: ratings.shape
```

```
Out[90]: (20000263, 3)
```

```
In [92]: ratings.isnull().any().any()
```

```
Out[92]: np.False_
```

```
In [93]: tags.shape
```

```
Out[93]: (465564, 3)
```

```
In [94]: tags.isnull().any().any()
```

```
Out[94]: np.True_
```

```
In [95]: tags = tags.dropna()
```

```
In [96]: tags.isnull().any().any()
```

```
Out[96]: np.False_
```

```
In [97]: tags.shape
```

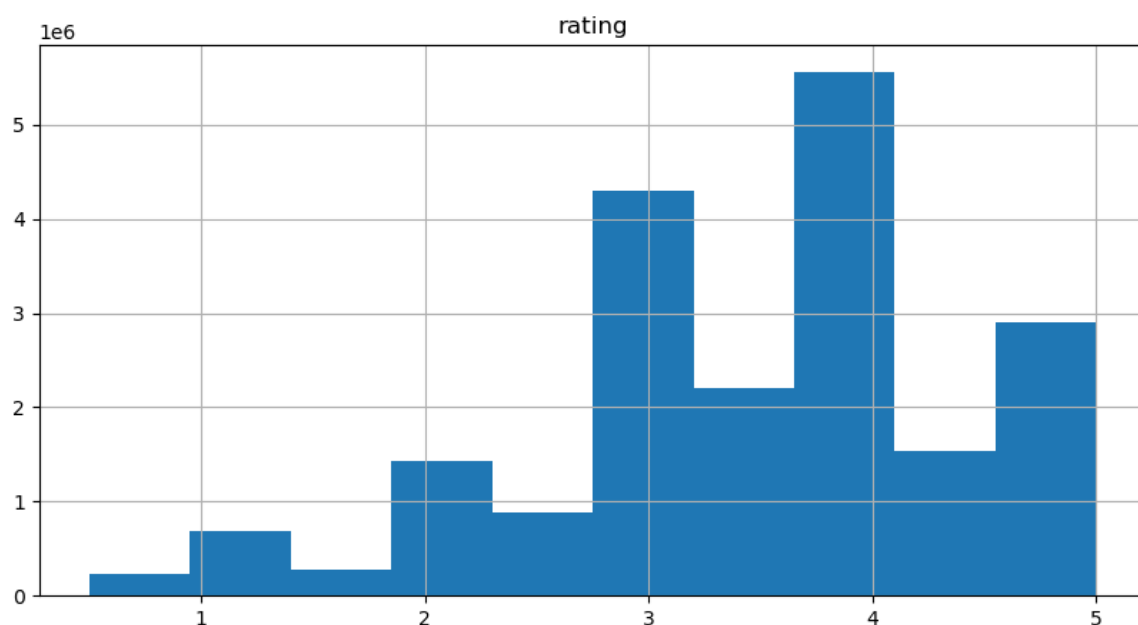
```
Out[97]: (465548, 3)
```

Data Visualization

```
In [98]: %matplotlib inline
```

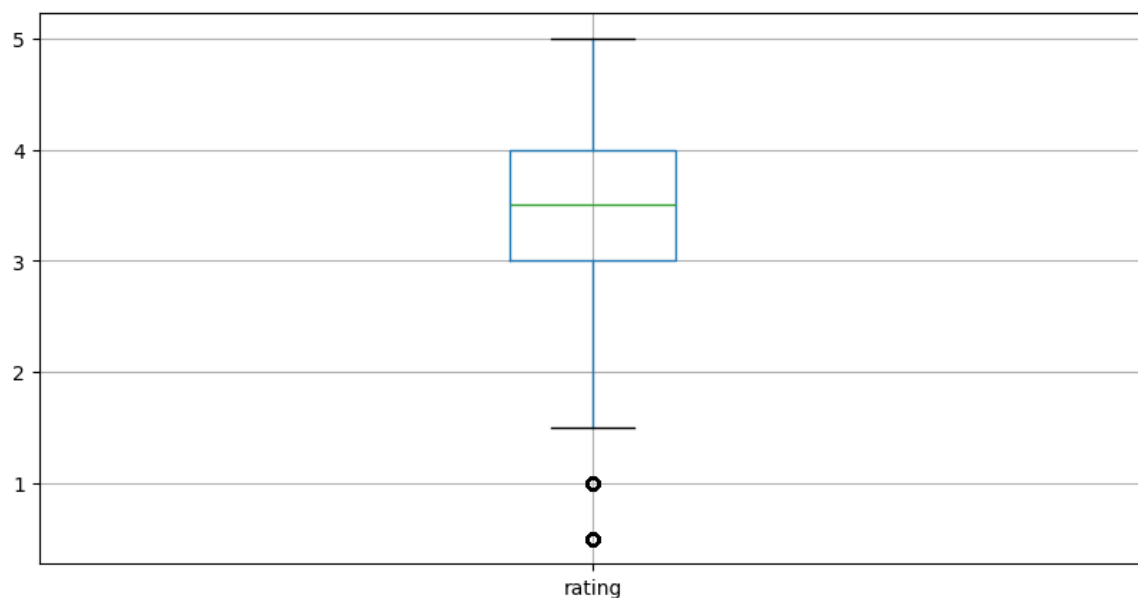
```
ratings.hist(column='rating', figsize=(10,5))
```

```
Out[98]: array([[<Axes: title={'center': 'rating'}>]], dtype=object)
```



```
In [99]: ratings.boxplot(column = 'rating',figsize =(10,5))
```

```
Out[99]: <Axes: >
```



Slicing Out Columns

```
In [100... tags['tag'].head()
```

```
Out[100... 0      Mark Waters
1      dark hero
2      dark hero
3      noir thriller
4      dark hero
Name: tag, dtype: object
```

```
In [102... movies[['title', 'genres']].head()
```

```
Out[102...
               title                                genres
0      Toy Story (1995)  Adventure|Animation|Children|Comedy|Fantasy
1      Jumanji (1995)      Adventure|Children|Fantasy
2      Grumpier Old Men (1995)      Comedy|Romance
3      Waiting to Exhale (1995)      Comedy|Drama|Romance
4      Father of the Bride Part II
      (1995)                                Comedy
```

```
In [103... ratings[-10:]
```

```
Out[103...
               userId  movieId  rating
20000253  138493      60816      4.5
20000254  138493      61160      4.0
20000255  138493      65682      4.5
20000256  138493      66762      4.5
20000257  138493      68319      4.5
20000258  138493      68954      4.5
20000259  138493      69526      4.5
20000260  138493      69644      3.0
20000261  138493      70286      5.0
20000262  138493      71619      2.5
```

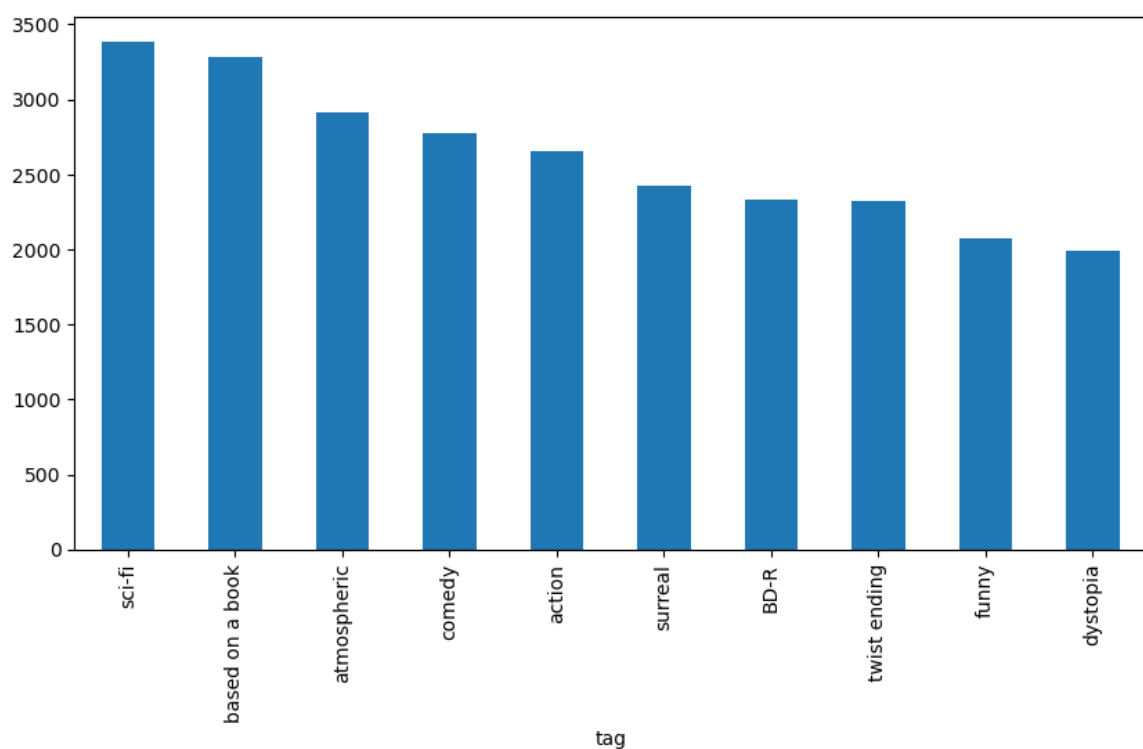
```
In [105... tag_counts = tags['tag'].value_counts()
tag_counts[-10:]
```



```
Out[105... tag
missing child          1
Ron Moore              1
Citizen Kane           1
mullet                1
biker gang             1
Paul Adelstein         1
the wig                1
killer fish            1
genetically modified monsters 1
topless scene          1
Name: count, dtype: int64
```

```
In [106... tag_counts[:10].plot(kind='bar', figsize=(10,5))
```

```
Out[106... <Axes: xlabel='tag'>
```



Filters for Selecting Rows

```
In [107... is_highlyRated = ratings['rating'] >= 5.0
ratings[is_highlyRated][30:50]
```

Out [107...

	userId	movieId	rating
239	3	50	5.0
242	3	175	5.0
244	3	223	5.0
245	3	260	5.0
246	3	316	5.0
247	3	318	5.0
248	3	329	5.0
252	3	457	5.0
253	3	480	5.0
254	3	490	5.0
256	3	541	5.0
258	3	593	5.0
263	3	858	5.0
264	3	904	5.0
267	3	924	5.0
268	3	953	5.0
271	3	1060	5.0
272	3	1073	5.0
275	3	1084	5.0
276	3	1089	5.0

In [108...

```
is_action= movies['genres'].str.contains('Action')  
movies[is_action][5:15]
```

Out [108...

movieId		title	genres
22	23	Assassins (1995)	Action Crime Thriller
41	42	Dead Presidents (1995)	Action Crime Drama
43	44	Mortal Kombat (1995)	Action Adventure Fantasy
50	51	Guardian Angel (1994)	Action Drama Thriller
65	66	Lawnmower Man 2: Beyond Cyberspace (1996)	Action Sci-Fi Thriller
69	70	From Dusk Till Dawn (1996)	Action Comedy Horror Thriller
70	71	Fair Game (1995)	Action
75	76	Screamers (1995)	Action Sci-Fi Thriller
77	78	Crossing Guard, The (1995)	Action Crime Drama Thriller
85	86	White Squall (1996)	Action Adventure Drama

In [109... `movies[is_action].head(15)`

Out [109...

movieId		title	genres
5	6	Heat (1995)	Action Crime Thriller
8	9	Sudden Death (1995)	Action
9	10	GoldenEye (1995)	Action Adventure Thriller
14	15	Cutthroat Island (1995)	Action Adventure Romance
19	20	Money Train (1995)	Action Comedy Crime Drama Thriller
22	23	Assassins (1995)	Action Crime Thriller
41	42	Dead Presidents (1995)	Action Crime Drama
43	44	Mortal Kombat (1995)	Action Adventure Fantasy
50	51	Guardian Angel (1994)	Action Drama Thriller
65	66	Lawnmower Man 2: Beyond Cyberspace (1996)	Action Sci-Fi Thriller
69	70	From Dusk Till Dawn (1996)	Action Comedy Horror Thriller
70	71	Fair Game (1995)	Action
75	76	Screamers (1995)	Action Sci-Fi Thriller
77	78	Crossing Guard, The (1995)	Action Crime Drama Thriller
85	86	White Squall (1996)	Action Adventure Drama

Group By and Aggregate

```
In [110... ratings_count = ratings[['movieId','rating']].groupby('rating').count()
ratings_count
```

```
Out[110...      movieId
rating
0.5      239125
1.0      680732
1.5      279252
2.0     1430997
2.5      883398
3.0     4291193
3.5     2200156
4.0     5561926
4.5     1534824
5.0     2898660
```

```
In [111... average_rating = ratings[['movieId','rating']].groupby('movieId').mean()
average_rating.head()
```

```
Out[111...      rating
movieId
1      3.921240
2      3.211977
3      3.151040
4      2.861393
5      3.064592
```

```
In [112... movie_count = ratings[['movieId','rating']].groupby('movieId').count()
movie_count.head()
```

Out [112...]

rating

movieId

	rating
1	49695
2	22243
3	12735
4	2756
5	12161

In []: